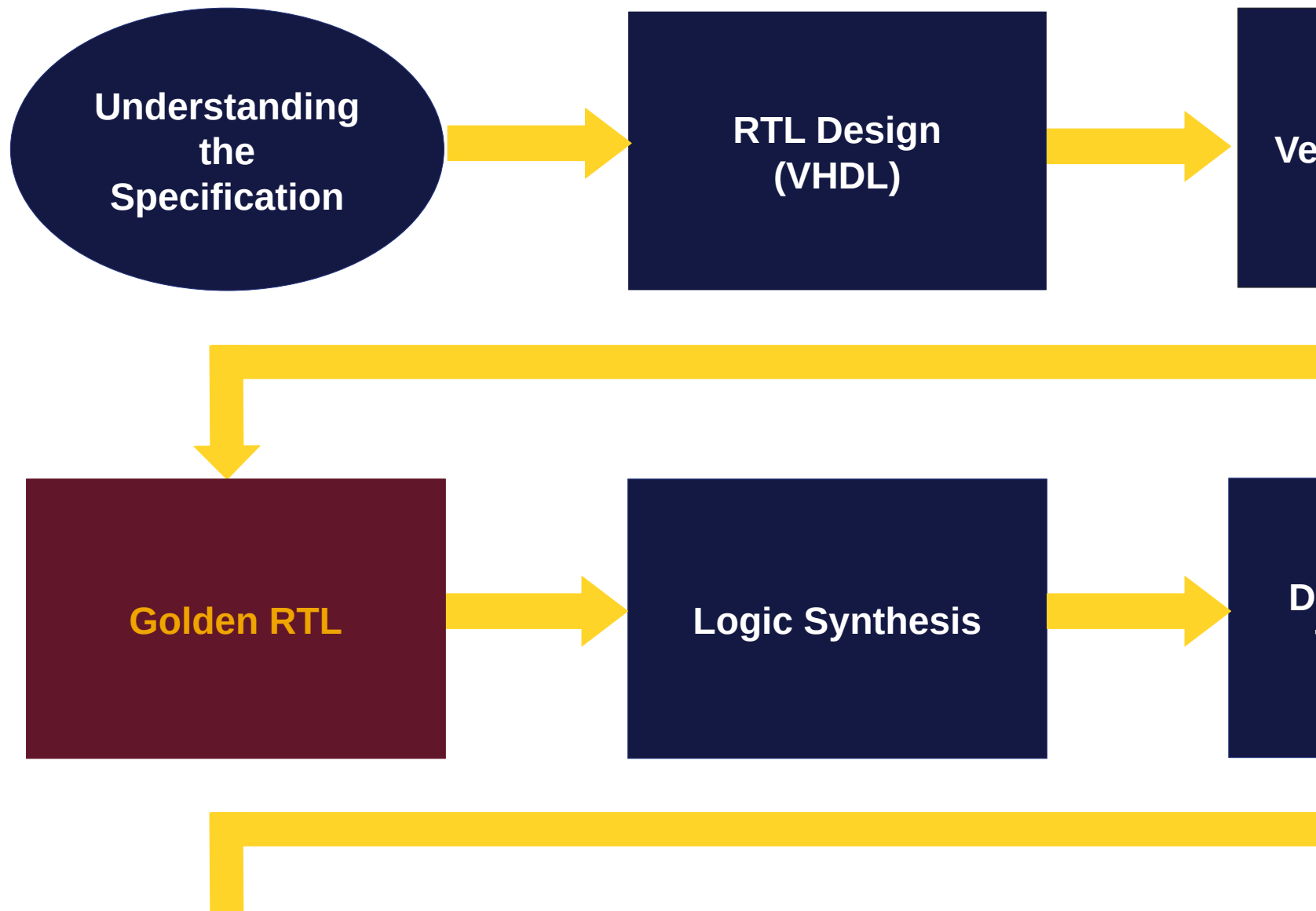


Development Memory-Interface AMBA-based Eth

ASIC Design Flow

From Specification to Final Design Tape-out



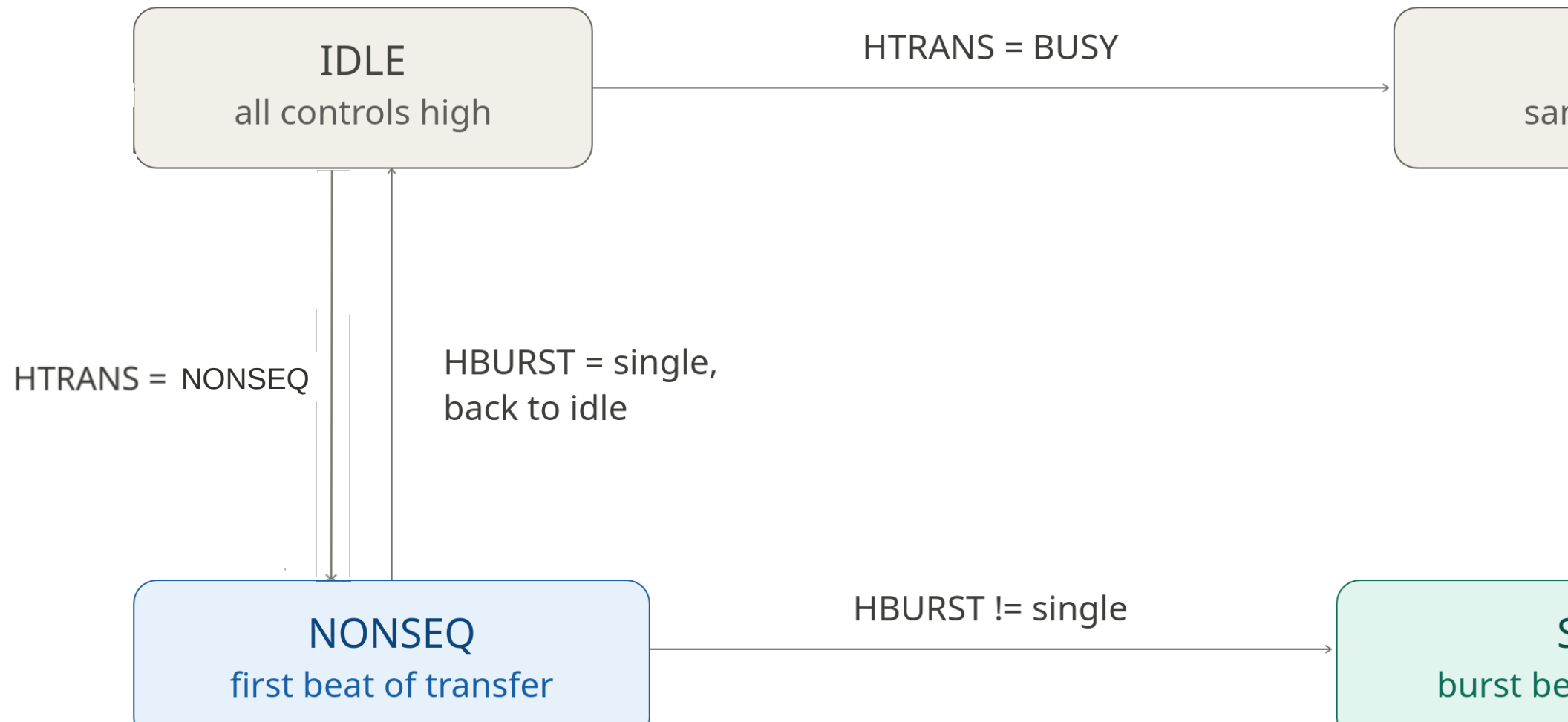
Design Block Diagram

Bridge Topology & Signal Grouping

- **AHB-Memory Bridge:** Converts AHB transactions from Ethernet MAC to SRAM read/write cycles
- **Memory-APB Bridge:** Provides processor access to Ethernet MAC configuration registers over APB
- **SRAM I/F:** Dedicated memory interface for TX/RX packet data
- **PROC I/F:** Processor-side Memory I/Os connected to APB-bridge

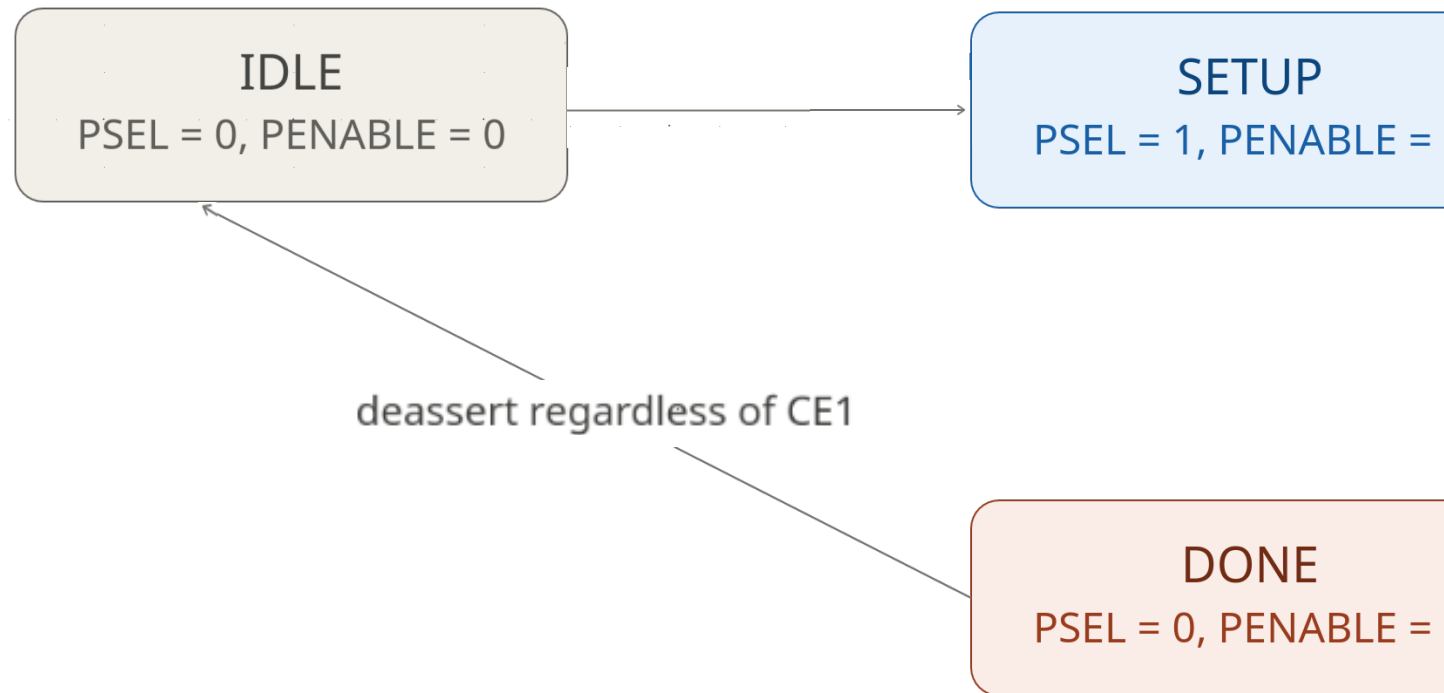
Design Ideation & FSMs

AMBA AHB: Control Signal Generation & BURST Handling Strategies



Design Ideation & FSMs

Transaction Phases & Control



pwrite drive logic (setup and access)

write: OE = 1, WE = 0 => PWRITE = 1

read: OE = 0, WE = 1 => PWRITE = 0

idle / done: PWRITE held at last driven value

Design Decisions & Justifications

Scope Boundaries & Limitations

- HREADY tied high — TACC = 10ns when
- Dedicated SRAM rationale — no arbitrated
SRAM isn't shared
- Burst support scope — supports all 8 Address
Modes of types: **SINGLE**, **INCR**, **WRAP**
- Scope boundary: HRESP tied to “OKAY”
other responses not yet implemented

Verification

LEON3 + GRETH MAC software verification, bridge testbench v



systest.c

Bare-metal C test on LEON3 + GRETH (5 tests)

Burst Transactions



GRETH Software Test

1

Register Access & MAC Address

Reset self-clear, MAC MSB/LSB write-readback, CTRL bit verification

2

PHY MDIO Read

Reads PHY ID1/ID2 and STATUS over the MDIO management interface

3

PHY Loopback Mode

Configures loopback MDIO write; confirm via readback

Findings & Corrections



MDIO Busy-Bit Mis-defined

MDIO_BUSY was coded as bit 0 instead of bit 3 — busy-wait exited immediately on the WRITE bit, corrupting MDIO transactions. Corrected to BUSY=bit3, READ=bit1, WRITE=bit0 per spec.



CTRL_PROM Bit Mis-positioned

Promiscuous-mode bit was defined as $(1 \ll 8)$ instead of the correct $(1 \ll 5)$ per the GRETH control register map. Corrected for spec compliance.

Code Coverage Analysis



Collection Method

```
vcs -cm line+cond+fsm+tgl+branch  
.simv -cm line+cond+fsm+tgl+branch  
urg -format both
```

VCS instrumented the RTL during the systest.c run; the urg Unified Report Generator produced the HTML/text coverage reports.

Metrics Collected:

- Line Coverage
- Condition Coverage

Expand All

NAME

[-] TEST

[-] CP

[-] PH

[-] PR

[-] PR

[-] PR

[-] PR

[-] SE

[-] SE

[-] SE

[-] SE

Standalone Testbench - Simple

Phase-1 bridge_top testbench · Basic connectivity & protocol sequencing · V0



Testbench Architecture

SRAM model

Combinational; ce1=0 & oen=0 → returns {addr[7:0], 0xABCDEF}

APB model

Simple echo: returns pwwdata as prdata when psel & penable

AHB stimulus

Driven manually — no hbusreq/hgrant/hready handshake

5

Test

addr

Test

addr

Test

0x4

Test

0x4

Test

3

StandAlone Testbench - Burst

Phase-2 bridge_top testbench · Full AHB protocol compliance · 19 test cases



Testbench Architecture

AHB Master Model

Full hbusreq/hgrant/hready handshake;
address+data pipelined 1 cycle; BUSY insertion;
early termination

Clocked SRAM

1024×32-bit array; registered addr; write/read by
ce1/oen/wen combination

Clocked APB

Te
Ca

P1

Wn

B1

SIN

B1

INC

E1

INC

Errors Found & Fixed

1

v_addr captured every cycle

Address grabbed from haddr on every clock, not just NONSEQ : double-incremented, skipping every other

2

count never reset between bursts

Reset only at exact burst limit; consecutive tests without full reset corrupted address/wrap

3

Address incremented by count, not +4

Increment used to beat counter its size — address just grew every beat

Lint Analysis using LEDA

- **Latch Generation :** LEDA detected potential latch inference
Solution: Added missing else branches.
- **Unregistered Top-Level Outputs:** Outputs were partially
Solution: Registered all applicable outputs except
- **Illegal Shorting of Input and Output Ports:** Direct port
Solution: Introduced intermediate registered signals
- **Clock-Gated Control Signal:** PENABLE was flagged as a
Solution: Redesigned logic to generate PENABLE
- **Incomplete Sensitivity Lists:** Missing dependencies were
Solution: Added required signals.

Design Compiler — Logic Syn

Padded bridge_top.vhd → gate-level netlist · 50 MHz constraint · Re-verified



Why a Padded RTL File?

Original bridge_top uses GRLIB record types (ahb_mst_out_type, apb_slv_in_type, etc.) requiring the full GRLIB library compiled to DC .db format.



Closed Verification Loop

Padded RTL re-verified using Simple_Test.vhd (all 5 tests) before submission to DC. Waveforms confirmed identical signal behaviour — no port mapping errors introduced during flattening.



Formality — Formal Equivalence

RTL (reference) vs. gate-level netlist (implementation) · Proves synthesis correct

What Formality Proves

For every possible input combination and current register state, the RTL source and the DC-generated netlist produce identical next-state and output values.

This guarantees that compile_ultra optimisations, technology mapping, and logic restructuring did not alter the functional behaviour of bridge_top.

Match Point Types:

Primary I/O

All ports — inputs & outputs of bridge_top

Formality (formal)

1. set
2. se
3. re
4. m
5. ve
6. D

Verific
ATTE
ATTE

Title

Caption

Title

Caption

Title

Caption

Thank

All questions are welcome